

Theoretical Computer Science Exam, July 10th, 2018

The exam consists of **4 exercises**. Available time: 1 hr 45 min.; you may write your answers in Italian, if you wish.

All exercises must be addressed for the written exam to be considered sufficient.

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I have passed the midterm exam and I intend to skip the first part (Ex.1-2) (select one): **YES NO**

Exercise 1 (9 pts)

Design minimal power automata that accept the following languages.

- 1) $\{ a^m b^n c^p d^q \mid m, n, p, q \geq 1 \text{ and } m + n = p + q \}$
- 2) $\{ a^m b^n c^p d^q \mid m, n, p, q \geq 1 \text{ and } m = p \text{ and } n = q \}$
- 3) $\{ a^m b^n c^p \mid m, n, p \geq 1 \text{ and } (m \text{ is odd if and only if } n \text{ is odd}) \text{ and } (\text{if } p \text{ is even then } n \text{ is even}) \}$

Exercise 2 (8 pts)

Write a Monadic First Order (or Monadic Second Order if necessary) logic formula that characterizes the language, over the alphabet $I = \{a, b\}$, whose strings are such that

1. all the a 's occur at positions with an odd index value (remember that in the word structure the index of position starts with the value 0), and
2. they include an odd number of a 's.

Exercise 3 (8 pts)

1. Consider the following function

$$h(x) = \text{if } f_x(x) = x \text{ then } 1 \text{ else } \perp$$

Is function h computable? Provide a clear justification for your answer.

2. (Optional) Consider the following set $S = \{ z \mid f_z(z) = z \}$. Is set S semidecidable? Justify your answer.

Exercise 4 (8 pts)

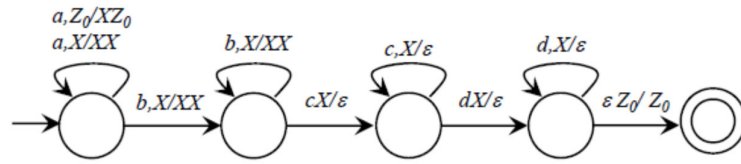
Consider the language, over the alphabet $I = \{a, b\}$, defined as follows $L = \{ aba^2b \dots a^k b \mid k \geq 1 \}$. In other words, the strings of L can be enumerated as follows: $L = \{ ab, abaab, abaabaaab, abaabaaabaaaab, abaabaaabaaaaab, \dots \}$.

Describe, in a precise and clear way, a Turing machine and a RAM machine that accept language L , preferably with minimal time complexity. Determine which of the two machines is more efficient, in a separate way for what concerns time and for what concerns space; suitably motivate your answers.

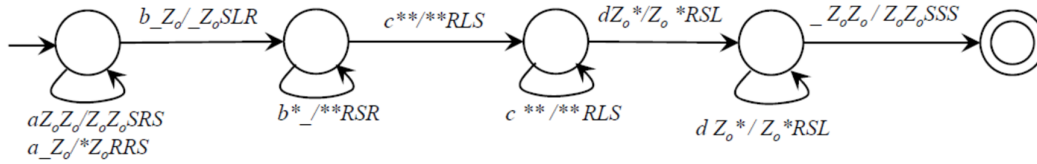
Solutions

Exercise 1

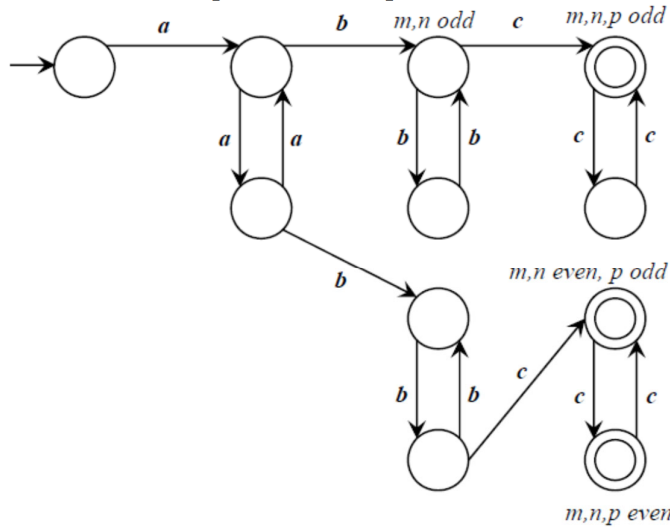
1. A simple PDA suffices



2. A TM is necessary



3. Since the constraints on the exponents concern their *parity*, a finite state automaton suffices. Notice that m and n must be either both odd or both even; furthermore, p can be even only if m and n are even, while p odd is compatible with any parity of m and n . Hence the allowed combinations of parity values for m, n, p , are: $\langle m, n, p \text{ odd} \rangle$, $\langle m, n \text{ even}, p \text{ odd} \rangle$, $\langle m, n, p \text{ even} \rangle$



Exercise 2

Predicate $N(x)$ means: number of a 's up to position x included is odd.

Predicate $X(x)$ means: position x is odd.

- 1.

$$\exists X \forall z (\neg X(0) \wedge (\neg \text{last}(z) \rightarrow (X(z) \leftrightarrow \neg X(z+1)) \wedge (a(z) \rightarrow X(z))))$$

- 2.

$$\begin{aligned} \exists N \forall x \forall y (& (\text{first}(x) \wedge a(x) \rightarrow N(x)) \wedge \\ & (\text{first}(x) \wedge \neg a(x) \rightarrow \neg N(x)) \wedge \\ & (S(x, y) \rightarrow ((N(x) \wedge a(y) \rightarrow \neg N(y)) \wedge \\ & \quad (N(x) \wedge \neg a(y) \rightarrow N(y)) \wedge \\ & \quad (\neg N(x) \wedge a(y) \rightarrow N(y)) \wedge \\ & \quad (\neg N(x) \wedge \neg a(y) \rightarrow \neg N(y))) \\ &) \end{aligned}$$

$$\begin{aligned} &) \wedge \\ & (\text{last}(x) \rightarrow N(x)) \\ &) \end{aligned}$$

Exercise 3

1. Function h is computable: a Turing machine can compute $h(x)$ by first simulating the computation of function $f_x(x)$ (function f_x is computable by definition); if the execution terminates and $f_x(x) = x$, output the value 1, otherwise go into a loop, therefore computing the undefined value $\perp$.
2. Yes, set S is semidecidable, precisely because function h in the previous point is computable.

Exercise 4

A one tape Turing machine can check the length of the segments of type $a^i b$ by storing in the memory tape sequences of symbols (say, ‘*’) of increasing length. The memory tape head can alternate scans from left to right and from right to left when it reads the successive segments of the input, so that it can analyze the input string in real time, while it needs an amount of memory which is equal to the length of the last segment (hence $\Theta(\sqrt{n})$, if $n = |x|$ is the length of the input).

The RAM machine can read the successive segments of the input, count their length, and check that they are of increasing length. With the uniform time criterion, the time complexity of the RAM is therefore almost real time, while the memory complexity is constant (a fixed number of variables storing integer values is needed). With the logarithmic cost criterion, the RAM machine is still superior to the Turing machine for what concerns space, but it is worse concerning time, because of the arithmetic computations of increasing cost that must be executed on the counters for each input symbol that is read from the input.